

Multilevel-based algorithm for hyperspectral image interpretation

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ARTICLE INFO

Keywords:

Hyperspectrum
Interpretation
Attention
Feature association
Semantic

ABSTRACT

Hyperspectral imagery contains spatial and spectral information, which can reveal the material properties of the target while intuitively displaying its spatial attributes. It has been applied in target recognition, search and rescue, and other fields. However, manual detection inevitably leads to missed detections and false alarms, necessitating the assistance of artificial intelligence for detection. To address this, we propose the multilevel-based algorithm for hyperspectral image interpretation. 1) From the spatial and spectral dimensions, we propose a semantic segmentation algorithm based on multidimensional information fusion to achieve semantic segmentation. 2) From the semantic and textual representation dimensions, we introduce a context interpretation module based on visual attention. We construct both real and simulated databases to validate the effectiveness of the algorithm. Experimental results demonstrate that the average accuracy of semantic segmentation achieved by the proposed algorithm is 74.3%. Additionally, the BLEU₁ score reaches 71.2, outperforming mainstream algorithms by 1.4.

1. Introduction

Hyperspectral imagery contains spatial and spectral information, resembling a 3D cube. The spectral curves corresponding to different ground objects exhibit variations, and the material information of objects can be reflected based on these curves. Currently, hyperspectral imagery has played a significant role in fire detection, object recognition, and camouflage detection [1,2].

Hyperspectral imagery can be regarded as a combination of multiple image frames. Scholars have conducted extensive research on how to extract valuable semantic information from these numerous images. Representative algorithms in this field include: Aguilar et al. [3] constructed local images and semantic relations to achieve semantic descriptions of images. Wang et al. [4] analyzed the images based on visual saliency. Imtiaz et al. [5] fused boundary features to an attention model to guide image description. Feng et al. [6] extracted hyperspectral semantic features based on wavelet transform. Qu et al. [7] combined attentional and LSTM models to describe images. Gan et al. [8] constructed semantic compositional networks to describe images. Wang et al. [9] implemented ground object recognition based on hyperspectral line features. Zhu et al. [10] constructed interpretation models based on contextual relations. Jiang et al. [11] applied a combination of long-term and short-term memory to image description. He et al. [12] focused on

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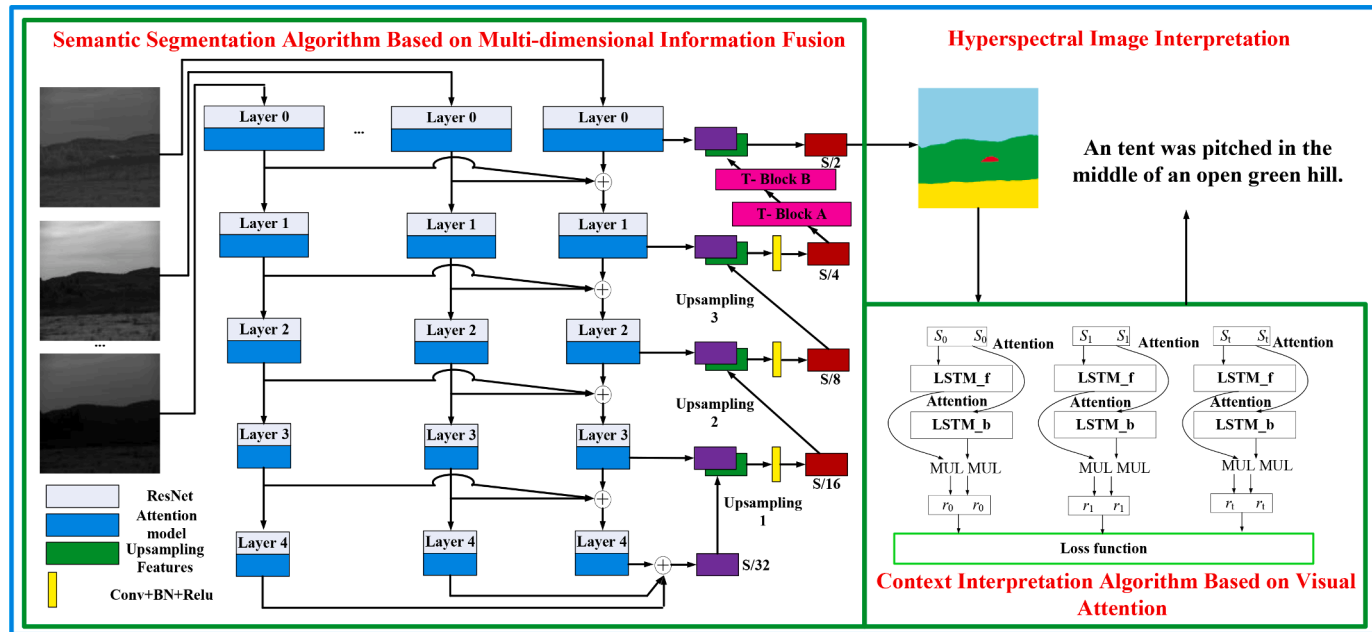


Fig. 1. Algorithm flow chart.

visual saliency regions to build a network for image interpretation. Donadello et al. [13] used a priori knowledge to guide accurate image description. Khowaja et al. [14] used video information to analyze target behavior for interpretation. Liu et al. [15] constructed an adaptive attention mechanism to interpret images. Zhang et al. [16] constructed a multimodal model for semantic analysis of images. Feng et al. [17] built double similarity distillation to achieve high-speed and high-precision semantic descriptions. Fan et al. [18] used self-attention neural architecture structure to implement semantic analysis. Mohamed et al. [19] carried out his research by modelling a convolutional dynamic auto-encoder based on a self-encoding structure. Lou et al. [20] fused lidar data with vision data to conduct 3D semantic research.

In conclusion, image interpretation is progressing towards intelligent development, and research based on deep networks has become a major approach. The main challenges faced in hyperspectral interpretation are as follows: 1) Insufficient utilization of spectral relationships in hyperspectral imagery leads to inaccurate semantic segmentation. 2) One-dimensional representations are limited in their ability to provide a global interpretation.

In light of these challenges, this paper conducts an in-depth investigation with the following innovations: 1) Proposal semantic segmentation algorithm based on multi-dimensional information fusion, implement semantic segmentation. 2) Proposal context interpretation algorithm based on visual attention, implement semantic segmentation.

The remainder of this paper is structured as follows: Section 2 presents the proposed algorithm. Section 2.1 presents semantic segmentation algorithm based on multi-dimensional information fusion. And Section 2.2 presents context interpretation algorithm based on visual attention. Section 3 presents experiment analysis. Section 3.1 presents segmentation algorithm performance. And Section 3.2 presents decoding algorithm performance. Section 4 presents conclusion offers conclusions from the main findings of this work with suggestions for future research.

2. The proposed algorithm

Hyperspectral imagery contains rich semantic information, and the complementarity and correlation of semantic information can effectively improve the accuracy of both the visual and semantic layers. To this end, we propose a multi-level hyperspectral image interpretation algorithm, as shown in Fig. 1. 1) The hyperspectral and spatial distribution characteristics of the target are fully exploited to construct a semantic segmentation algorithm based on multi-dimensional information fusion, achieving semantic segmentation. 2) According to the visual perceptual process, the attention mechanism is established to explore contextual information. Based on semantic features, target predicates and relational predicates are classified, and attention mechanism features are integrated into an LSTM model to generate descriptive sentences.

2.1. Semantic segmentation algorithm based on multi-dimensional information fusion

Hyperspectral imagery is composed of multiple spectral bands, each containing distinct information. To fully leverage the advantages of hyperspectral data, we propose a semantic segmentation model consisting of M encoders and 1 decoder. Both the encoder and decoder consist 5 layers. A lightweight attention module is incorporated after each layer to extract diverse information. During the fusion stage, the feature maps from each branch are summed to generate the output.

The encoder and decoder consist of three main modules: the upsampling module, feature map concatenation module, and feature extraction module. As the depth of the encoding network increases, high-level semantic features are captured. However, these high-level semantic features capture global contextual information at the expense of losing fine-grained image details. The decoder employs upsampling computations, which result a lack of detailed information and produce blurry predicted boundaries. To enhance the clarity of the output predicted boundary map, a lightweight attention module is introduced to guide the encoder's downsampling process towards emphasizing the predicted features from multiple bands. In order to simultaneously consider both high-level semantic information and detail information, the upsampling feature maps are concatenated with the corresponding-sized feature maps from the encoder stages using the concatenation module. These concatenated feature maps are then passed through convolutional layers to extract features, enabling the decoder to balance semantic and detail information while continuously restoring resolution. This approach benefits the final output of the predicted boundary map, thereby optimizing the effectiveness of semantic segmentation.

The encoder backbone network is composed of ResNet [21,22], and its corresponding lightweight attention module consists of

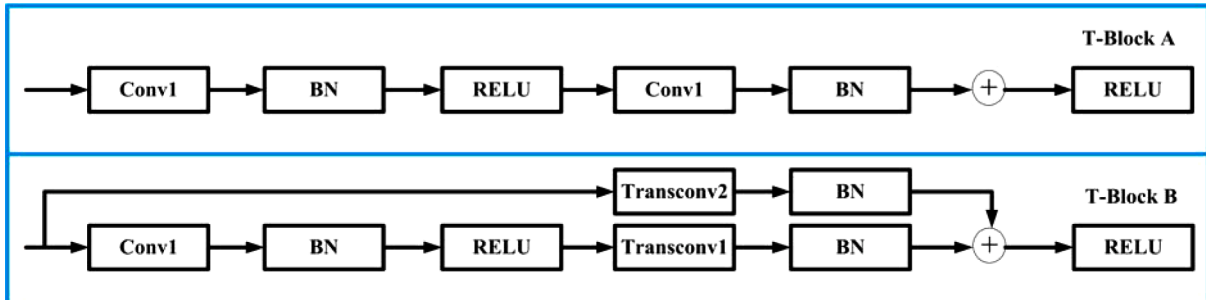


Fig. 2. Schematic diagram of convolutional block operation.

channel attention and spatial attention. The channel attention focuses on global features, while the spatial attention emphasizes global regions. The channel attention enables local cross-channel interaction strategy, where the coverage range k of local cross-channel interaction is determined by the number of channels C . The relation between the two can be expressed as

$$k = \left\lceil \frac{\log_2 C}{\gamma} + b \right\rceil \quad (1)$$

where $\lceil t \rceil$ represents the nearest even number to t . γ and b are hyperparameters.

The decoder operation diagram is shown in Fig. 2, where the resolution and channel numbers of the feature maps remain consistent with the input after the first module. In the second module, Conv1 maintains the resolution of the output data while reducing the channel number to half of its original value. Subsequently, Transconv1 increases the resolution of the feature maps by a factor of two while keeping the channel number unchanged, whereas Transconv2 halves the channel number and doubles the resolution of the feature maps.

The encoder combines weighted fusion of feature maps from different bands to form S_0 . In the decoder, the feature map S_1 is first upsampled to match the channel number and resolution of S_2 . S_1 and S_2 are then concatenated and fused to obtain S_3 . The feature extraction module is utilized to generate the feature map S_4 by iteratively extracting features. Finally, a softmax layer is applied to obtain the probability map for segmentation results. By fusing the feature maps containing high-level semantic information and low-level detail information, and employing multi-scale techniques, the boundaries of the resulting semantic segmentation map are made clearer, leading to improved segmentation performance.

2.2. Context interpretation algorithm based on visual attention

Based on the semantic segmentation results of the images, CNN is employed to detect and extract a set of candidate regions, denoted as B . The number of regions of interest (ROI) is controlled using non-maximum suppression, and the targets are connected to form a relationship graph. A scene graph, denoted as $G=\{B,O,R\}$, is established to represent the structural semantics of the images. B represents the set of candidate bounding boxes, O represents the set of targets, and R represents the relationships between the targets.

Based on the ROI, a spatial relationship connection graph is established, which is considered as subject-predicate-object and connected to two target proposals. Given that GNN can retain the intrinsic features of nodes and accept fused representations, we employ GNN for facilitating the propagation and update of node information. Within a single time step, hidden layers are incorporated to aggregate the previous state of nodes and information from neighboring nodes in order to update each node representation. The target region information primarily serves the recognition of object relationships, with features extracted from the target being used as inputs for the relationship nodes. The descriptive region mainly encompasses information regarding the relationships between targets. Features are extracted from both the target region and the relationship region to serve as inputs for the descriptive nodes. Based on the information propagation graph, we construct representations of object nodes and relationship nodes:

$$m_{is}^t = \left[\sum_{j \in N_i^s} (f_j^s, h_j^{t-1}), \sum_{j \in N_i^r} (f_j^r, h_j^{t-1}) \right] \quad (2)$$

$$m_{io}^t = \left[\sum_{j \in N_i^o} (f_j^s, h_j^{t-1}), \sum_{j \in N_i^r} (f_j^r, h_j^{t-1}) \right] \quad (3)$$

where at moment t , m_{is}^t and m_{io}^t are the information received by the subject and the object respectively. f_j^s and f_j^o are subject and object node information. f_j^r indicates the relation node information. N_i^s and N_i^o are the set of nodes adjacent to the subject and object. The update process is as:

$$\begin{cases} z_i^t = \sigma(W_o^s m_i^t + U_o^s h_i^{t-1}), r_i^t = \sigma(W_o^r m_i^t + U_o^r h_i^{t-1}) \\ \tilde{h}_i^t = \tanh(W_o m_i^t + U_o(r_i^t \odot h_i^{t-1})), h_i^t = (1 - z_i^t) \odot h_i^{t-1} + z_i^t \odot \tilde{h}_i^t \end{cases} \quad (4)$$

where σ is the activation function, $\odot$ is the vector dot product. Nodes in the graph aggregate information about neighbouring nodes and propagate their own information to other nodes. The node information is propagated through the graph and undergoes T iterations to obtain the final hidden state $\{h_1^T, h_2^T, \dots, h_N^T\}$ of the region corresponding to each target node, and a fully connected network $f_i^o = \phi_o(h_i^o; h_i^T)$ is constructed. The relation for predicting the class of objects is established as shown below:

$$l_i^o = \operatorname{argmax}[\delta_o(f_1^o, f_2^o, \dots, f_N^o)] \quad (5)$$

The classification of targets enables the detection of relationships between them. We initialize object nodes using feature vectors from target regions and relationship nodes using feature vectors from joint regions composed of subject and object target regions, along with spatial information of the targets.

$$\begin{cases} h_o = \varphi_o(f_i^s, f_j^o) \\ h_r = \varphi_r(f_{ij}^r, s_j^s, s_j^o) \end{cases} \quad (6)$$

where φ_o and φ_r are the transformations of the fully connected layer, f_i^s and f_j^o are the feature vectors of the areas corresponding to the subject and object objects, and f_{ij}^r denotes the feature vector of the joint area formed by the subject and object target areas. s_j^s, s_j^o are the spatial information of the subject and object targets. h_o is the hidden state of the target nodes and h_r is the hidden state of the relation nodes. The corresponding relation process is as:

$$m_{ir}^t = \left[\sum_{j \in N_i^s} (f_j^s, h_j^{t-1}), \sum_{j \in N_i^o} (f_j^o, h_j^{t-1}) \right], \begin{cases} f_i^r = \phi_r(h_r^0, h_i^{T_r}) \\ l^r = \arg\max [\delta_r(f_1^r, f_2^r, \dots, f_M^r)] \end{cases} \quad (7)$$

Each descriptive region node in the information propagation graph is connected to multiple target and relationship nodes. We describe the nodes using feature vectors composed of target and relationship regions.

$$h_c = \varphi_c \left[\sum_{i=1, j \in N^s}^N (f_i^s, f_j^o), \sum_{i=1}^M f_i^r \right] \quad (8)$$

where φ_c denotes a transformation at the fully connected layer and h_c denotes the hidden state of the nodes. For describing the transfer of updates to node information, the relation equation is constructed as:

$$\begin{cases} m_{ic}^t = \left[\sum_{j \in N_i^s} (f_j^s, f_j^o, h_i^{t-1}), \sum_{i=1}^M (f_j^r, h_j^{t-1}) \right] \\ f_i^c = \phi_c(h_c^0, h_i^{T_c}) \end{cases} \quad (9)$$

By utilizing the aforementioned feature information propagation structure, features from different semantic levels are aggregated from connected nodes, allowing for the updating of individual features through the process of propagating information to neighboring nodes.

The algorithm incorporates visual attention units, semantic attention units, and fusion units. The generated structural model is illustrated in Fig. 3. The visual attention units are responsible for obtaining the weights α for each image region at the current time step. The semantic attention units determine the probability β of preserving textual information at the current time step. The fusion units first generate visual context C^V based on α and the global image feature vector V , and semantic context C^T based on g and the LSTM storage unit m . The visual-semantic context C^0 is obtained by fusing C^V , C^T , and β . This attention mechanism considers historically attended regions and introduces a textual preservation probability, allowing the model to autonomously decide whether to focus on the image or the generated text. The weight calculation is denoted as

$$e_t = \begin{cases} W_0 V_t = 0 \\ W_v V + W_h h_{t-1} + e_{t-1} + bt > 0 \end{cases} \quad (10)$$

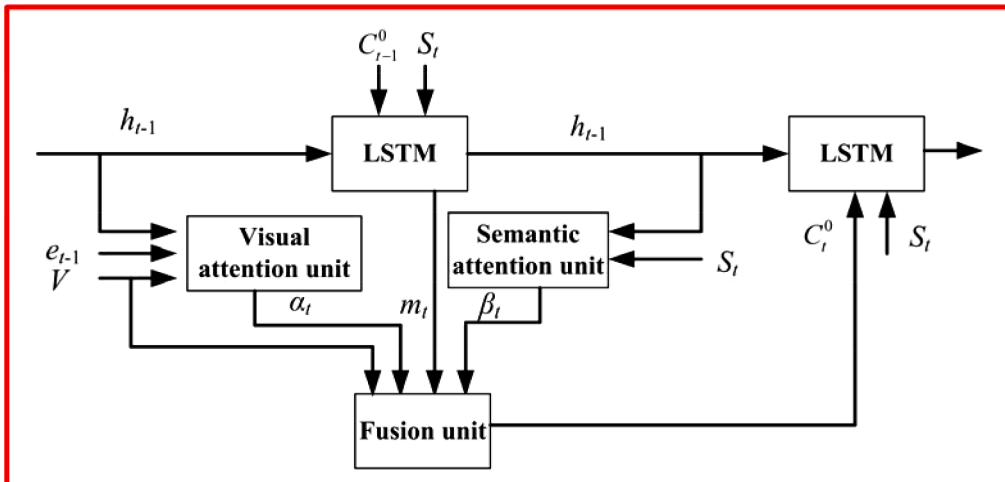


Fig. 3. Structural model.

$$\alpha_t = \text{softmax}(W_e e_t) \quad (11)$$

where e_t denotes the weight's intermediate state of the focused area at moment t and can be considered as the memory unit. Both W and b can be obtained by training. When $t = 0$, the hidden state h_0 and the storage cell state c_0 of the LSTM are determined by the average global features of the input images. The formula for calculating the probability of retaining text generated by the semantic attention unit at moment t is as follows:

$$\beta_t = \text{softmax}[W_e(W_s S_t + W_h h_t)] \quad (12)$$

The visual context C_t^V and the semantic context C_t^T of the fusion unit at moment t are calculated as:

$$C_t^V = \sum_{i=1}^L \alpha_i^j V_i, \begin{cases} \alpha_i^j = \exp(e_i^j) / \sum_{k=1}^L \exp(e_i^k) \\ e_i^j = f_{att}(V_i, h_{t-1}) \end{cases} \quad (13)$$

$$\begin{cases} C_t^T = g_t \tanh(m_t) \\ g_t = \sigma(W_x S_t + W_h h_{t-1} + W_c C_{t-1}^0) \\ C_t^0 = \beta C_t^T + (1 - \beta) C_t^V \end{cases} \quad (14)$$

The introduction of the attention mechanism allows the visual attention storage unit to retain areas of historical attention, and the semantic attention module to autonomously choose to focus on previously generated text, making the model more accurate for each focused area at every moment.

The text generation module consists of a multimodal fusion layer that predicts the text corresponding to the images based on the input word S , the hidden state h of the LSTM, the visual semantic context and a triad of visual attributes. A two-layer LSTM is used to generate text from 2 directions. During training, the weights are updated according to the sum of the forward and backward prediction losses. During testing, the model output is selected by the maximum value of the probability sum of the forward and backward generated text.

- 1) Compute the encoding vector for the input word. Let $S = \{s_0, \dots, s_b, \dots, s_n\}$. Assuming that the input word at time t is w_t , the relation between w_t and S is constructed using word embedding to reduce the dimensionality.
- 2) Update the LSTM hidden state. The update formula is as:

$$\begin{cases} f_t = \sigma(W_f[h_{t-1}, S_t, C_{t-1}^0] + b_f), m_t = f_t m_{t-1} + i_t m_i \\ i_t = \sigma(W_i[h_{t-1}, S_t, C_{t-1}^0] + b_i), o_t = \sigma(W_o[h_{t-1}, S_t, C_{t-1}^0] + b_o) \\ m_i = \tanh(W_c[h_{t-1}, S_t, C_{t-1}^0] + b_c), h_t = o_t \tanh(m_t) \end{cases} \quad (15)$$

where m_t is the cell state at moment t and σ denotes the sigmoid function. f_t denotes the forget gate function, which is used to control the information retained by the cell at the migration moment. i_t represents the input gate function for updating the current moment information. o_t denotes the output gate function, which controls the output of the updated cell state. W and b correspond to the parameters respectively.

- 1) Construct the visual triad, and the feature vector is obtained as:

$$r_t = W_h h_t + S_t + W_c C_t^0 + W_{attr} V_{attr} \quad (16)$$

where W is a learnable parameter. The feature vector y is obtained through the fully connected layer and corresponds to the following objective function:

$$\theta^* = \underset{\theta}{\operatorname{argmax}} \sum_{I, S} \log p(S|I; \theta) \quad (17)$$

where S denotes the calibrated description of the image I . The loss function of the training process is as:

$$\begin{cases} L(I, S) = Q(\vec{w}_t) + Q(\overleftarrow{\vec{w}_t}) + \lambda \|\theta\|^2 \\ Q(a_t) = - \sum_{t=0}^N \sum_{t=0}^T \log(p(a_t|I, V_{attr}, a_0, \dots, a_{t-1})) \end{cases} \quad (18)$$

where $Q(\vec{w}_t)$ and $Q(\overleftarrow{w}_t)$ are the cross-entropy loss functions for the forward and backward LSTM, N is the training sample image size, and w_t denotes the word generated at moment t . The text is generated according to the forward LSTM and backward LSTM and the maximum probability formula for the sentence is obtained as:

$$p(w_{1:T}|I, V_{attr}) = \max \left(\sum_{t=1}^T p(\vec{w}_t|I, V_{attr}), \sum_{t=1}^T p(\overleftarrow{w}_t|I, V_{attr}) \right) \quad (19)$$

3. Experiment analysis

The experiment utilized a total of 10,000 sets of hyperspectral images, consisting of both real captured and software-simulated datasets, illustrated in Fig. 4. The images were annotated to identify the targets present within them. The training and testing data were maintained a ratio of 1:1. The experimental program was developed using VS2010, and the Windows operating system served as the execution platform. To assess the effectiveness of the proposed algorithm, we established the following metrics for evaluation.

3.1. Segmentation algorithm performance

We introduce ACC (Accuracy) and IOU (Intersection over Union) to evaluate the segmentation effect, where ACC denotes the ratio of correctly predicted pixels to the total number of pixels, and IOU denotes the intersection of the true labels and the predicted results for each category [23]. In order to reflect the effect of the model more intuitively, the $mACC$ (mean Accuracy) and $mIOU$ (mean Intersection over Union) are usually calculated as:

$$mACC = \frac{1}{N} \sum_{i=1}^N \frac{n_{ii}}{\sum_{j=1}^N n_{ij}} \quad (20)$$

$$mIOU = \frac{1}{N} \sum_{i=1}^N \frac{n_{ii}}{\sum_{j=1}^N n_{ij} + \sum_{j=1}^N n_{ji} + n_{ii}} \quad (21)$$

where n_{ij} denotes the pixel point with true category i and predicted as category j .

The proposed Multi-dimensional information fusion semantic segmentation algorithm presented in this paper is compared with existing mainstream algorithms, as shown in Table 1. The Low-Level Feature Enhancement Network (LLFEN) [24] utilizes the intrinsic properties of objects to enhance the low-level texture features for semantic segmentation. However, the semantic segmentation

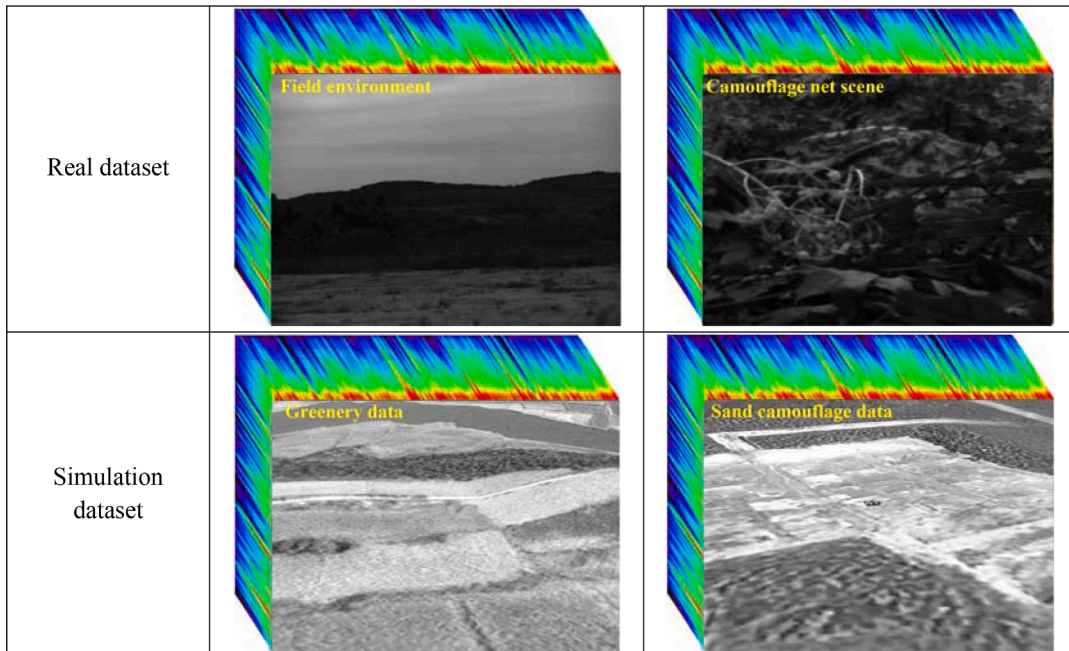


Fig. 4. Dataset presentation.

Table 1
Segmentation effect.

Algorithm	LLFEN	FuseSeg	GFCN	CENet	GMNet	Ours
<i>mACC</i> (%)	59.1	62.1	66.4	69.3	70.5	74.3
<i>mIoU</i> (%)	44.2	45.7	49.6	51.2	54.7	57.1

performance using only low-level features is unsatisfactory. The Thermal Data Fusion Network (FuseSeg) [25] improves the semantic segmentation performance by extracting and analyzing features from multiple channels of data. The Graph-FCN (GFCN) [26] considers both local and global features, transforming the semantic segmentation problem into a graph node classification problem. It applies graph convolutional networks to solve this classification problem, further enhancing the semantic segmentation performance. The Contextual Ensemble Network (CENet) [27] captures context through ensemble deconvolution, obtaining multi-scale feature maps and achieving good semantic segmentation results. The Graded-feature Multilabel-learning Network (GMNet) [28] proposes a hierarchical feature extraction strategy that targets different regions of the image more effectively, divides the multi-level features into primary, intermediate, and advanced levels, and optimizes the network using multi-label supervision in terms of semantic, binary, and boundary features to achieve semantic segmentation. The proposed information fusion semantic segmentation algorithm in this paper, the spatial information and spectral information are integrated to construct the model with good effect, resulting in the best performance with an *mACC* of 74.3%, surpassing mainstream algorithms by 3.8%.

For further analysis of the algorithm's performance, we selected large and small targets as representatives and plotted their ROC curves as shown in Fig. 5. Experiments were conducted on both real and simulated datasets. A comparison between (Fig. 5(a) and Fig. 5(b)) and (Fig. 5(c) and Fig. 5(d)) reveals that the algorithm performs better in segmenting large targets than small targets. Contrasting Fig. 5(a) and Fig. 5(c) indicates that, under the premise of segmenting large targets, the algorithm's performance on the simulated dataset surpasses that on the real dataset. Contrasting Fig. 5(b) and Fig. 5(d) reveals that, under the premise of segmenting small targets, the algorithm's performance on the real dataset surpasses that on the simulated dataset. This is attributed to the lower resolution of the simulated dataset and the presence of internal smoothing methods such as filtering, which reduces the intra-class differences of large targets and weakens the information of small targets.

The semantic segmentation results corresponding to the algorithm proposed in this paper are shown in Fig. 6. In the face of real collected hyperspectral images, Fig. 6(a) depicts a field environment during dusk, including the sky, land, greenery, and tents. It is difficult to distinguish the difference between the tents and the greenery from the RGB image. However, the algorithm proposed in this paper effectively achieves accurate discrimination. Fig. 6(b) presents data containing a camouflage net scene, where the camouflage

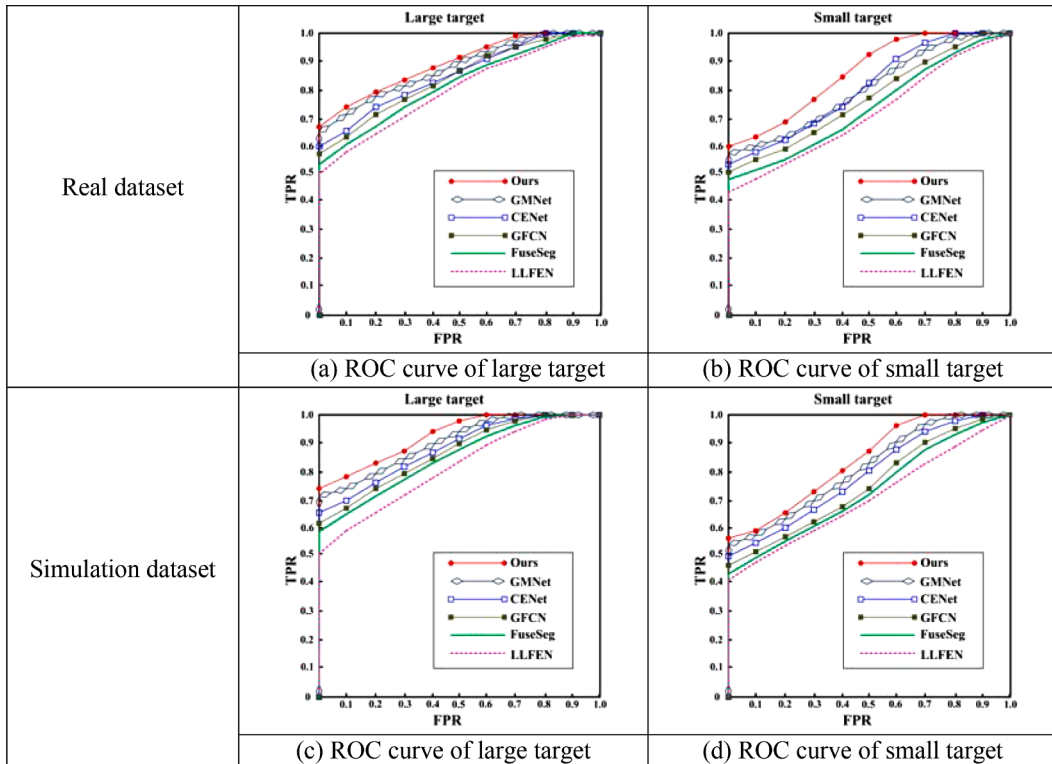


Fig. 5. ROC curve.

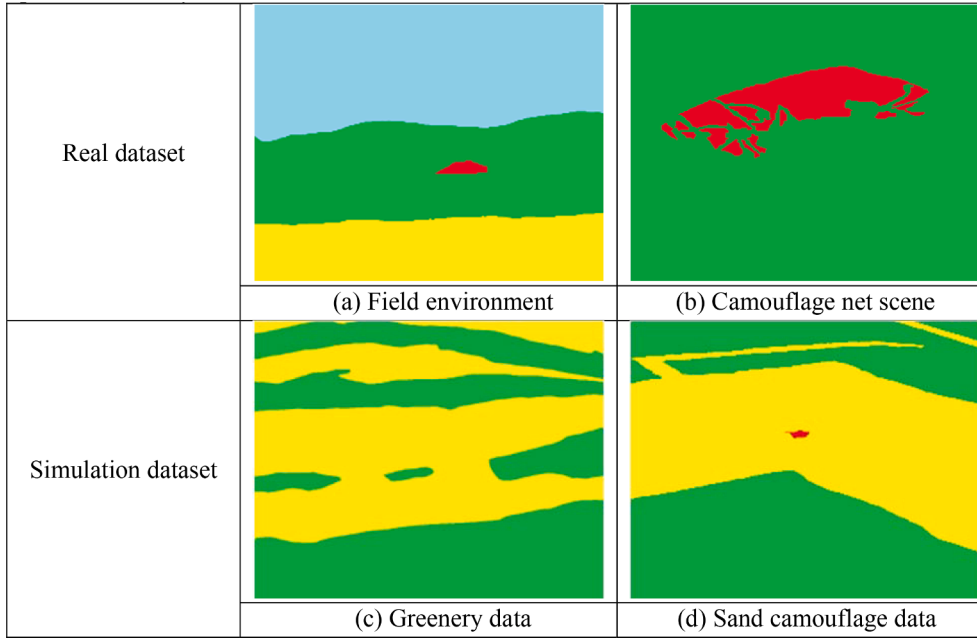


Fig. 6. Semantic segmentation results.

net is obstructed by vegetation. Nevertheless, the proposed algorithm still performs well. The simulated generated hyperspectral image in terms of spatial resolution exhibits a block-like pattern. Fig. 6(c) represents the greenery data, encompassing land and grass, with an overall satisfactory segmentation result. Due to limitations in resolution, there are cases where some land portions are incorrectly labeled as grass in the upper right corner. Fig. 6(d) illustrates sand camouflage data, with a tank parked in the center. It is challenging to differentiate between the desert and the tank from the RGB image alone, but the algorithm proposed effectively labels them.

Based on the above analysis, the paper thoroughly explores spatial and spectral dimensional information, constructing a semantic segmentation algorithm with information fusion, achieving significant effectiveness. When dealing with low-resolution hyperspectral data, there is a possibility of missegmentation.

3.2. Decoding algorithm performance

The generated sentences were evaluated using *BLEU* and *METEOR* [29]. *BLEU* measures the degree of co-occurrence between candidate sentences and reference sentences in terms of n-grams. *METEOR* considers both precision and recall of the entire corpus, showing a high correlation with human judgments. Both *BLEU* and *METEOR* are positively correlated with the quality of semantic understanding.

$$\begin{cases} BLEU_N = BP \exp \left(\sum_{n=1}^N w_n \log p_n \right) \\ BP = \begin{cases} 1, & \text{if } c > r \\ \exp(1 - r/c), & \text{other} \end{cases} \end{cases} \quad N = \{1, 2, 3, 4\} \quad (22)$$

where c represents the length of the generated sentence, r represents the length of the reference sentence, p_n stands for the precision of modified n-grams, and w_n stands for weight.

Table 2

Generated text effects.

Algorithm	$BLEU_1$	$BLEU_2$	$BLEU_3$	$BLEU_4$	<i>METEOR</i>
Attr-Without	62.3	43.5	31.6	21.1	19.0
Attr-Traditional	67.4	47.4	35.5	25.5	21.1
Attr-Triple	68.8	49.7	37.8	28.6	22.3

$$\begin{cases} METEOR = M(1 - Q) \\ Q = 0.5 \left(\frac{C_{num}}{U_{num}} \right)^3 \\ M = \frac{10PR}{R + 9P} \end{cases} \quad (23)$$

where P and R represent the precision and recall of unigram segmentation. Considering the factor of sentence length, a penalty term Q is designed. C_{num} is the number of chunks, and U_{num} is the number of matched unigrams.

We employed visual attribute triplets (Attr-Triple) (<object, action, scene>), traditional visual attribute extraction (Attr-Traditional), and none-attribute implementation (Attr-Without) for image semantic interpretation. As shown in Table 2, the visual attribute model achieved favorable results. The model performed even better when applied to the dataset with visual attribute triplets. Models without attribute features may lead to incorrect predictions of image themes based solely on text. This indirectly demonstrates the effectiveness of the visual attribute model in reducing the probability of errors. Traditional attribute prediction often involves words of limited length, which may have overlapping or semantically similar meanings, resulting in redundant attributes. In contrast, visual attribute triplets accurately predict the main attributes of an image while reducing the complexity of model training by reducing the number of parameters.

With a batch size of 64 and 15,000 iterations, the proposed Attr-Triple model's loss variation is depicted in Fig. 7. As the number of iterations increases, the curve demonstrates a decreasing trend and eventually stabilizes, indicating that the model is trainable and effective. The curves illustrating the changes in *BLEU* and *METEOR* evaluation scores are presented in Fig. 8. Both metrics exhibit an upward trend, reaching stability at around 10,000 iterations. The model proposed in this study successfully identifies the main objects and attribute information in the images, selectively attends to either the image or previously generated text, and combines future information to generate diverse and contextually rich image descriptions, thereby achieving semantic understanding of the images. However, due to the large volume of hyperspectral data, the network we constructed takes all the hyperspectral data as input, resulting in long training and testing times. Subsequently, we can explore band selection algorithms to optimize the data, thereby improving the algorithm's performance.

The algorithm proposed in this paper is compared with mainstream algorithms as shown in Table 3. DeepVS [3] associates multiple local regions in the image with textual description segments and generates textual descriptions of the image using an RNN. GoogleNIC [30] is a CNN and LSTM-based image semantic description model that reduces the number of model parameters. The Hard-Attention [4] model calculates attention weights for each local region, allowing it to focus on smaller regions at each time step. VA-LSTM [7] establishes six color dictionaries, predicts object boundaries, incorporates color and contour information of the image in the attention mechanism, and generates textual descriptions. Saliency+Context+Att (SCA) [31] utilizes SAM to obtain saliency maps of the image, and then fuses the saliency map with global image features to generate corresponding text using an LSTM. CNN-RNN-f3 [32] incorporates textual feedback information into the traditional CNN-RNN structure, reinforcing the attentional focus through a recurrent iterative structure. LSTM-EM-DA [11] introduces storage units and gate signals to the input and output gates of the LSTM, respectively, and incorporates attention from the previous time step to determine the current attended region. ATT-FCN [5] combines image visual attributes, LSTM, and word decoding algorithms. SCN-LSTM [8] extends an LSTM parameter for each visual attribute, assigning a probability to each attribute, and calculates the generated word at each time step based on weighted visual attributes. LSTM+ATT+CNN [33] modularizes the image feature extraction, attribute prediction, sentence feature extraction, and text generation modules. The input to the text generation module includes the previously generated word, predicted attributes, and contextual sentence features. VD-BLSTM [12] proposes a shared-parameter image semantic understanding model that utilizes DenseNet for image feature extraction, visual attribute prediction, and incorporates attention models into LSTM for interpretation. In summary, both DeepVS and GoogleNIC only input image information at the beginning of the text generation module, and do not incorporate image-related data afterwards. As a result, as time progresses, the effective information in the image decreases, leading to low model performance and accuracy. Other algorithms construct models from a single perspective and fail to fully extract the hierarchical features of high-spectral imagery. In contrast, the proposed algorithm in this paper can locate specific regions and combine deep image features for image understanding. By considering the temporal basis of attending to previously generated text and focused regions, the model is able to connect historical and future information, resulting in more accurate localization and generation of diverse and rich sentences, effectively improving the accuracy of image semantic understanding.

To intuitively demonstrate the interpretation performance, we present the interpretation results of the proposed algorithm in this paper. 1) Field environment: A tent was pitched in the middle of an open green hill. 2) Camouflage net scene: A camouflage color tent was hidden among a pile of trees. 3) Greenery data: In an open field there was regularly planted some vegetation. 4) Sand camouflage data: A tank moves across an exposed field surrounded by green vegetation. It is evident that the algorithm proposed in this paper achieves a good interpretation performance by leveraging the spatial and material information of the targets.

4. Conclusion

Hyperspectral imagery can reflect the spatial and material information of targets and can be widely applied to target detection and emergency rescue. Faced with the challenge of hyperspectral interpretation, we have developed a multilevel interpretation algorithm. Firstly, we construct real and simulated datasets. Then, by fully exploiting the spatial and spectral information of hyperspectral data, we propose a semantic segmentation algorithm based on multidimensional information fusion. Finally, simulating the visual

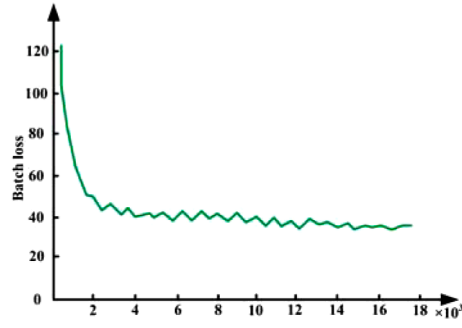


Fig. 7. Training variation curve.

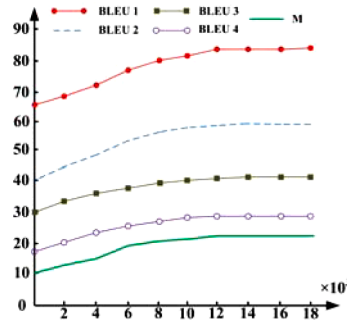


Fig. 8. Iterations and performance tests.

Table 3

Generated text effects.

Algorithm	$BLEU_1$	$BLEU_2$	$BLEU_3$	$BLEU_4$	METEOR
Deep VS	58.6	37.1	23.1	15.8	14.9
Google NIC	59.8	38.2	24.5	16.9	—
Hard-Attention	60.1	39.4	26.3	17.3	15.6
VA-LSTM	61.3	40.1	26.8	17.9	—
SCA	61.9	41.3	27.6	18.6	16.7
CNN-RNN-f3	62.3	43.2	27.9	20.3	17.3
LSTM-EM-DA	65.4	44.5	29.3	21.6	17.9
ATT-FCN	66.8	46.7	30.4	22.9	18.1
SCN-LSTM	67.1	48.9	33.8	25.6	19.2
LSTP+ATT+CNN	68.2	49.1	35.6	26.4	20.5
VT-BLSTM	68.8	49.7	37.8	28.6	22.3
Ours	71.2	50.1	38.5	29.3	25.1

perception process, we implement hyperspectral data interpretation through a context interpretation algorithm based on visual attention. However, the current research is limited to existing hyperspectral data collection and simulated data, and all hyperspectral image data are input into the network, resulting in long processing times. Subsequent efforts are needed to collect hyperspectral data from different devices and conduct research on band selection to further study the efficiency and robustness of the algorithm.

Author statement

We would like to submit the enclosed manuscript entitled “Multilevel-Based Algorithm for Hyperspectral Image Interpretation”, which we wish to be considered for publication in “Computers and Electrical Engineering”. No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication.

Declaration of Competing Interest

Multilevel-Based Algorithm for Hyperspectral Image Interpretation. The authors declare that they have no conflict of interest.

Data availability

Data will be made available on request.

Acknowledgments

This work is supported by Supported by Light of West China (No. XAB2022YN10), the National Key Research and Development Program of China (No. 2019YFE0126500), The National Natural Science Foundation of China (No. 41971359), the Scientific Instrument Developing Project of the Chinese Academy of Sciences, (No. YJKYYQ20200010).

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